

Connect and Protect: Networks and Network Security

Module 1

Network architecture

Get started with the course

- ✓ Introduction to Course 3
Video • 1 min
- ✓ Course 3 overview
Reading • 4 min
- ✓ Helpful resources and tips
Reading • 4 min

Introduction to networks

- ✓ Welcome to module 1
Video • 39 sec
- ✓ Chris: My path to cybersecurity
Video • 3 min
- ✓ What are networks?
Video • 1 min
- ✓ Tina: Working in network security
Video • 2 min
- ✓ Emmanuel: Useful skills for network security
Video • 2 min
- ✓ Network tools
Video • 2 min
- ✓ Network components, devices, and diagrams
Reading • 4 min
- ✓ Practice: Design a local area network
Ungraded Plugin • 10 min
- ✓ Cloud networks
Video • 2 min
- ✓ Cloud computing and software-defined networks
Reading • 8 min
- ✓ Identify: Functions of network tools
Ungraded Plugin • 10 min
- ✓ Test your knowledge: Introduction to networks
Practice Assignment • Grade: 100%

Network communication

- ✓ Introduction to network communication
Video • 2 min
- ✓ The TCP/IP model
Video • 2 min
- ✓ The four layers of the TCP/IP model
Video • 1 min
- ✓ Learn more about the TCP/IP model
Reading • 8 min
- ✓ The OSI model
Reading • 8 min
- ✓ Test your knowledge: Network communication
Practice Assignment • Grade: 100%

Local and wide network communication

- ✓ IP addresses and network communication
Video • 2 min
- ✓ Components of network layer communication
Reading • 4 min
- ✓ Test your knowledge: Local and wide network communication
Practice Assignment • Grade: 100%

Review: Network architecture

- ✓ Wrap-up
Video • 39 sec
- Glossary terms from module 1
Reading • 4 min
- Module challenge 1
Graded Assignment • 50 min

Module 2

Glossary terms from module 1

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Terms and definitions from Course 3, Module 1

Bandwidth: The maximum data transmission capacity over a network, measured by bits per second

Cloud computing: The practice of using remote servers, application, and network services that are hosted on the internet instead of on local physical devices

Cloud network: A collection of servers or computers that stores resources and data in remote data centers that can be accessed via the internet

Data packet: A basic unit of information that travels from one device to another within a network

Hub: A network device that broadcasts information to every device on the network

Internet Protocol (IP): A set of standards used for routing and addressing data packets as they travel between devices on a network

Internet Protocol (IP) address: A unique string of characters that identifies the location of a device on the internet

Local Area Network (LAN): A network that spans small areas like an office building, a school, or a home

Media Access Control (MAC) address: A unique alphanumeric identifier that is assigned to each physical device on a network

Modem: A device that connects your router to the internet and brings internet access to the LAN

Network: A group of connected devices

Open systems interconnection (OSI) model: A standardized concept that describes the seven layers computers use to communicate and send data over the network

Packet sniffing: The practice of capturing and inspecting data packets across a network

Port: A software-based location that organizes the sending and receiving of data between devices on a network

Router: A network device that connects multiple networks together

Speed: The rate at which a device sends and receives data, measured by bits per second

Switch: A device that makes connections between specific devices on a network by sending and receiving data between them

TCP/IP model: A framework used to visualize how data is organized and transmitted across a network

Transmission Control Protocol (TCP): An internet communication protocol that allows two devices to form a connection and stream data

User Datagram Protocol (UDP): A connectionless protocol that does not establish a connection between devices before transmissions

Wide Area Network (WAN): A network that spans a large geographic area like a city, state, or country

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